

Read Pico Power Management

read_pico_pmu datasheet · firmware 1.0.8

Read Pico is an ESP32–S3 development board with a 4.7" monochrome e–paper panel, made by Shenzhen MindReset Technology Co., Ltd. for developers who want to write open–source firmware for e–paper devices. This document is about the power–management chip on that board: how it sequences power, how it talks to the host, and what the host should do.

The chip is programmed once. There is no OTA / IAP. Production parts enable Flash readout protection level 1 through ISP.

Item	Value
Chip	read_pico_pmu
Company	Shenzhen MindReset Technology Co., Ltd.
Board	Read Pico (小纸 Pico)
Firmware	1.0.8
Protocol	1.1
Build fingerprint	20260905
device_id	0x3201
I2C slave address	0x2A (7–bit)
Silicon	Wuhan Xinyuan CW32L010F8U6 (QFN20, 3.0 × 3.0 mm)
Date	2026–09–17

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1. Overview

read_pico_pmu stays powered on the board. It handles power, battery, charge state, and the key. The ESP32-S3 host runs only after the PMU drives `ESP_EN` high. They talk over I2C; the PMU is the slave. Events go out on open-drain `CW_INT` (active low).

1.1 Roles

Role	Responsibility
read_pico_pmu	Power sequencing, key decisions, battery ADC and SOC, charge-pin decode, LED policy, IWDT, RTC and alarm, maintenance data and VCOM
ESP32-S3	Application, UI, filesystem, power-action requests, event consumption, time sync, panel VCOM read/write

Charge detection, voltage conversion, and key debounce are done on the PMU. The host reads the results. Command `OK` / `ACCEPTED` only means the PMU accepted the request. Power state is `STATUS.power_state`.

1.2 Features

- 9-state power machine: boot, cooperative shutdown, forced off, hard reset, soft sleep, download
- Two-phase dangerous actions (PREPARE token ~2 s + COMMIT, delay cap 5 s)
- Battery ADC: internal bandgap for VDDA, 33 kΩ / 100 kΩ divider, SOC table (permille, monotonic while not charging)
- Thresholds: low 3500 mV, critical 3300 mV, forced off 3050 mV, hysteresis 100 mV
- LY4566-420 charge pin: blink = charging, steady = full
- Short / long / 8 s event / 10 s forced off / double-short then long for ESP32 download
- Event FIFO depth 8, PEEK + ACK one by one, CRC16
- On-chip RTC (LSI): UTC sync, one-shot/repeat alarm 10 s–31 days, can power the host on
- DeepSleep after 15 s idle while off and not charging (µA class)
- Last Flash page: equivalent cycles, replace count, repair codes, panel VCOM
- Session `boot_id` + (session_id, sequence) idempotency cache of 4
- No OTA; production readout lock via official ISP

1.3 Block diagram

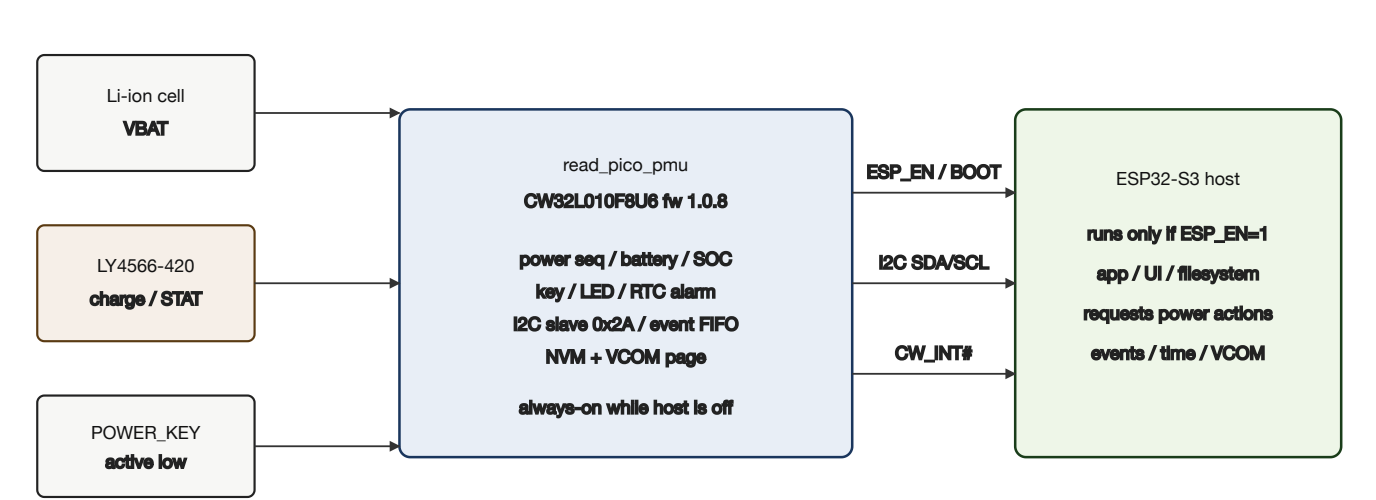


Figure 1-1. PMU is always powered; the host runs only when `ESP_EN=1`.

1.4 Identity

After power-up the host must read IDENTITY (register 0x00 , 32 bytes).

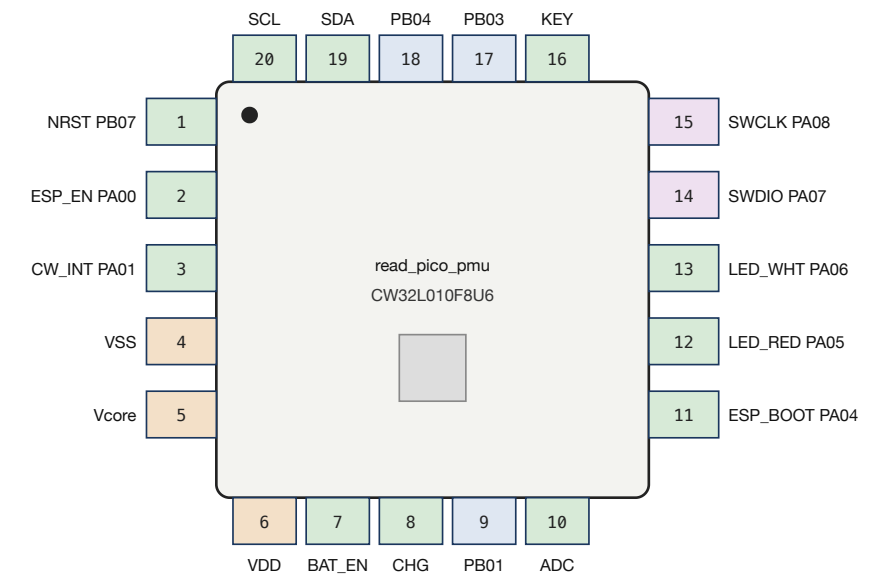
Item	Value
Magic	ASCII PMU1
Protocol	1.1
Firmware	1.0.8 (fw_major/minor/patch)
hw_revision	1
device_id	0x3201
Capabilities	bits 0–5: BATTERY_ADC, CHARGE_STATUS, KEY_EVENTS, LED_CONTROL, HOST_EN_CONTROL, HOST_BOOT_CONTROL
Max frame	64 bytes
Event FIFO	8
build_number	low 16 bits of 20260905 (0x4F09)
build_hash	20260905

boot_id is regenerated on every PMU reset. When it changes, drop old sequence / token and re-read IDENTITY, STATUS, and events. A USB reset of the ESP32 alone does not change boot_id .

2. Pinout

Package is CW32L010F8U6, QFN20, 3.0 × 3.0 mm. The table uses this firmware’s net names. Unused mux functions are in the vendor *CW32L010 datasheet* table 5–2.

QFN20 top view (3.0 x 3.0 mm) ? read_pico_pmu



green = firmware used orange = power purple = SWD/ISP gray = unused
Top view, counterclockwise. Dot = pin 1. Vcore: 100-470 nF to GND, do not supply externally.
7=PA02 BAT_MEASURE_EN 8=PB00 CHARGE_STATUS 10=PA03 BAT_ADC 16=PB02 POWER_KEY 19=PB05 20=PB06

Figure 2-1. QFN20 top view (counter-clockwise, dot is pin 1). Green = used, orange = power, purple = SWD/ISP, gray = unused.

2.1 Pins

QFN	Silicon	Net	Dir	Description
1	PB07 / NRST	NRST	I	Chip reset, internal pull-up
2	PA00	ESP_EN	PP out	Host enable, active high. Held low on cold start
3	PA01	CW_INT	OD out	Event IRQ, low = FIFO not empty. Host must pull up
4	VSS	GND	Power	Ground
5	Vcore	Vcore	Power	Internal regulator. 100–470 nF to ground. Do not supply externally
6	VDD	CW_VDD	Power	Chip supply, ~3.3 V on this board
7	PA02	BAT_MEASURE_EN	PP out	Enables the battery divider (N–MOS gate)
8	PB00	CHARGE_STATUS	In	LY4566–420 STAT. Bleeded before sample
9	PB01	—	—	Unused
10	PA03	BAT_ADC	Analog	Battery divider sample
11	PA04	ESP_BOOT	PP out	Host BOOT / IO0. High normally; held low then EN pulsed for download
12	PA05	LED_RED	Out / PWM	Red LED
13	PA06	LED_WHITE	Out / PWM	White LED
14	PA07 / SWDIO	SWDIO	Debug / ISP	Same pad as UART1_RXD. SWD after reset, internal pull-up
15	PA08 / SWCLK	SWCLK	Debug / ISP	Same pad as UART1_TXD. SWD after reset, internal pull-up
16	PB02	POWER_KEY	In	Power key, active low
17	PB03	—	—	Unused
18	PB04	—	—	Unused
19	PB05	I2C_SDA	OD	I2C SDA
20	PB06	I2C_SCL	OD	I2C SCL

2.2 Debug and ISP

PA07 / PA08 are SWD after reset. For production readout lock over ISP, the same pads become UART1: hold reset → feed ~50 kHz on SWDIO → release reset and wait 5 ms. No extra ISP pads are required.

3. Electrical characteristics

Unless noted, numbers are from the CW32L010 datasheet Rev 1.0 ($T_A = 25\text{ }^{\circ}\text{C}$, $V_{DD} = 3.3\text{ V}$) and this firmware's operating point. System clock is internal HSI 48 MHz / 6 = **8 MHz**. No external crystal.

3.1 Absolute maximum ratings (vendor)

Parameter	Symbol	Min	Max	Unit
Main supply	$V_{DD} - V_{SS}$	−0.3	6.0	V
IO input	V_{IN}	$V_{SS} - 0.3$	$V_{DD} + 0.3$	V

Parameter	Symbol	Min	Max	Unit
Total sink on one VDD			100	mA

Exceeding these ratings can cause permanent damage.

3.2 Recommended operating conditions

Parameter	Min	Typ	Max	Unit	Note
VDD	1.62	3.3	5.5	V	3.3 V LDO on this board
Operating temperature	−40		85	°C	
System clock		8	48	MHz	This firmware uses 8 MHz; 48 MHz needs VDD ≥ 1.8 V
I2C rate	100	400	400	kHz	Slave runs at 400 kHz
GPIO VIH	0.7 VDD			V	
GPIO VIL			0.3 VDD	V	
GPIO source/sink			±8	mA	Up to ±20 mA if VOL/VOH is relaxed

3.3 Power (vendor typical, VDD = 3.3 V)

Mode	Condition	Typ	Unit
Run, code in Flash, peripherals off	HSI 8 MHz	1.7	mA
Sleep, code in Flash/RAM	HSI 8 MHz, peripherals off	0.45	mA
DeepSleep	Regulator on, oscillators off	0.3	µA
DeepSleep	Regulator on, LSE/RTC/IWDT on	1.2	µA

I2C responds while the host is running or charging. DeepSleep starts after 15 s idle while off and not charging. IWDT is paused in DeepSleep. Board-level DeepSleep current also includes charger leakage and the divider path; measure on the board. Expect µA class.

3.4 Memory and clocks

Item	Value
Flash	64 KB. Last 512 B page (0xFE00) is maintenance data
SRAM	4 KB
Firmware size	~31 KB code + 0.5 KB maintenance page
High-speed clock	Internal HSIOSC, DIV6 → 8 MHz
Low-speed clock	Internal LSI (~32 kHz): IWDT, RTC
RTC accuracy	About ±1–2%, tens of minutes per day. Host must TIME_SYNC regularly

3.5 Timing constants

Symbol	Value	Meaning
KEY_DEBOUNCE	30 ms	Key debounce
KEY_POWER_ON	1000 ms	Long-press power-on while EN=0
KEY_LONG	1500 ms	Report KEY_LONG
KEY_FORCE_EVENT	8000 ms	Report KEY_FORCE_OFF (does not cut power)

Symbol	Value	Meaning
KEY_FORCE_OFF	10000 ms	Forced power-off while EN=1
KEY_COMBO_GAP	1200 ms	Max gap between the two shorts
KEY_DOWNLOAD_HOLD	3000 ms	Third long-press enters download
ESP_RESET_PULSE	250 ms	Hard-reset EN low pulse
ESP_EN_HOLD	20 ms	EN low hold on the download path
ESP_BOOT_SETUP	10 ms	Hold BOOT low before pulsing EN
ESP_BOOT_RELEASE	500 ms	Release BOOT after entering download
SHUTDOWN/SOFT_SLEEP EN delay	100 ms	Update STATUS, then drop EN
HOST_BOOT_TIMEOUT	15 s	Wait for HOST_READY; timeout only sets a fault bit
SHUTDOWN_TIMEOUT	10 s	Cooperative shutdown limit
ACTION_TOKEN_TTL	2 s	PREPARE token lifetime
ACTION_DELAY_MAX	5 s	Max delay after COMMIT
DEEPSLEEP_IDLE	15 s	Idle window before DeepSleep
I2C_STUCK	100 ms	Slave self-recovers after a stuck bus
IWDT	1 s	Overflow reset; paused in DeepSleep
LED_OVERRIDE	5 s	Default LED_SET override time
LED fade default / max	400 / 4000 ms	

4. Typical application

The PMU is powered from the always-on 3.3 V rail. Host 3.3 V is gated by `ESP_EN`.

4.1 Battery divider

Upper 33 k Ω , lower 100 k Ω (ratio 1.33). `BAT_MEASURE_EN` drives an N-MOS to close the path. Wait 10 ms, then sample. The path is off otherwise to cut leakage. Valid range 2700–4400 mV.

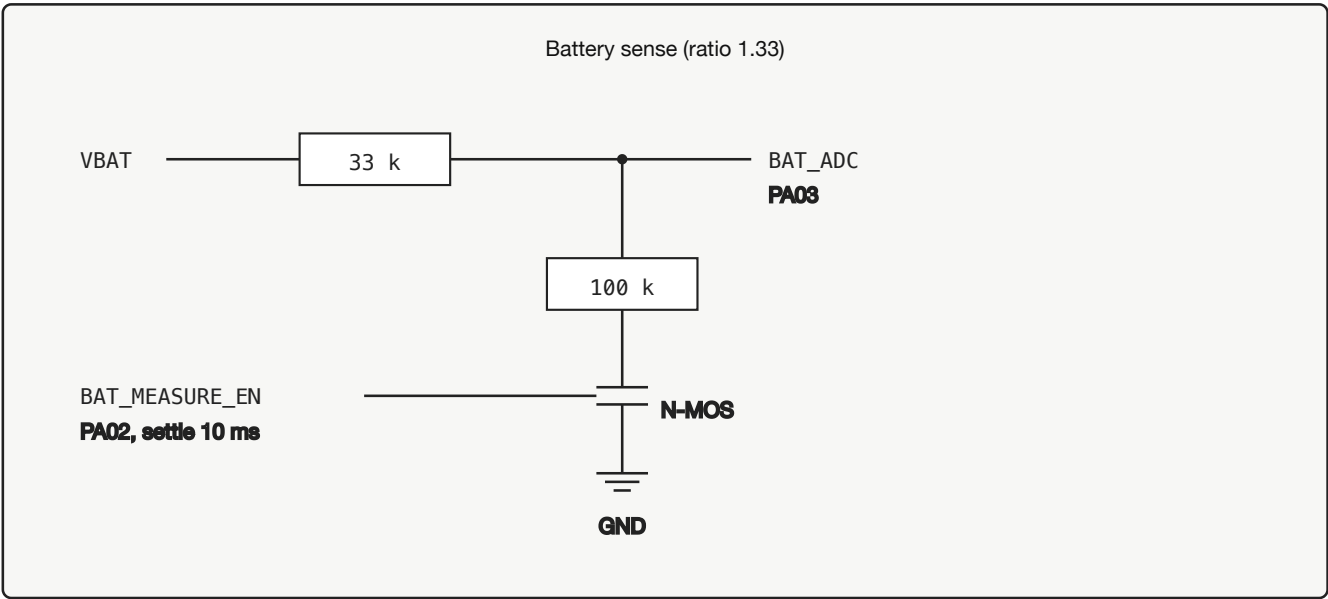


Figure 4-1. Battery divider. 33 kΩ / 100 kΩ, N-MOS gated by BAT_MEASURE_EN.

4.2 Charge indication

Charger is LY4566–420. STAT goes to PMU `CHARGE_STATUS` : blink while charging (low half-period about 730–800 ms on this board), steady when full, low when unplugged.

4.3 Host interface

Signal	Host side	Note
ESP_EN	EN / enable	High = powered
ESP_BOOT	GPIO0	Sampled for download
SDA / SCL	I2C master	4.7–10 kΩ pull-up to host 3.3 V
CW_INT	GPIO input, pulled up	Low = unacked events
POWER_KEY	—	Wired to the PMU only

Host I2C may share the bus with other slaves. Access the PMU serially inside one task. Do not split a frame.

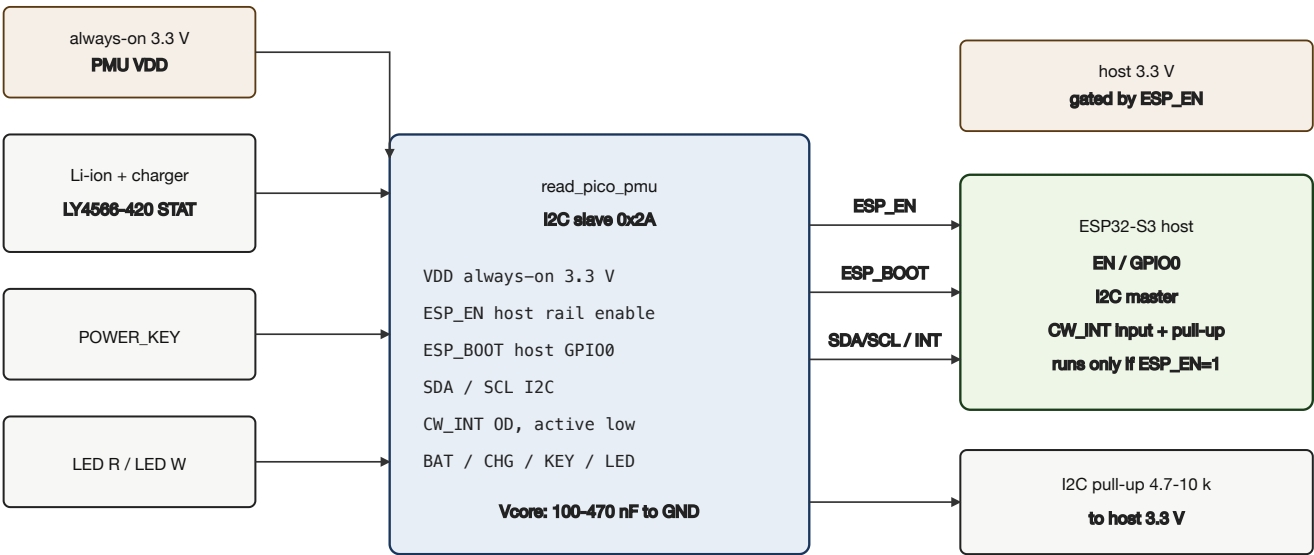


Figure 4-2. Power and host connections. Net names follow chapter 2.

5. Power state machine

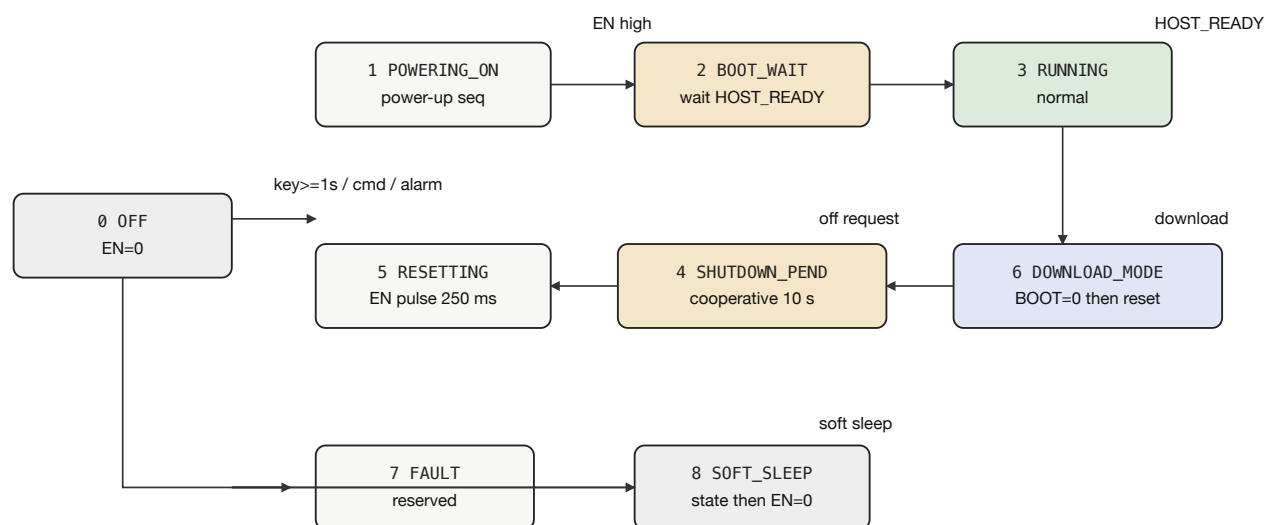


Figure 5-1. Command completion is STATUS.power_state, not the command return code.

Value	Name	EN	Meaning
0	OFF	0	Host unpowered. Cold-start default. May DeepSleep
1	POWERING_ON	Rising	Power-up sequence (brief)
2	BOOT_WAIT	1	Waiting for HOST_READY. After 15 s only <code>fault_flags</code> bit0 is set; power stays on
3	RUNNING	1	Normal
4	SHUTDOWN_PENDING	1	Cooperative shutdown, wait SHUTDOWN_READY, force off at 10 s
5	RESETTING	Pulse	Hard reset: EN low $\geq$ 250 ms, then BOOT_WAIT
6	DOWNLOAD_MODE	1 after pulse	BOOT=0 then reset. Release BOOT after ~500 ms. HOST_READY returns to RUNNING
7	FAULT	—	Reserved; not entered
8	SOFT_SLEEP	1 then 0	STATUS first, EN=0 after ~100 ms. Next short press only raises EN

5.1 Two-phase actions

Use ACTION_PREPARE (0x0300) + ACTION_COMMIT (0x0301).

1. PREPARE: `action:u8 + delay_ms:u16 + reason:u16`. Returns `token:u32 + expires_in_ms:u16 (2000) + effective_delay_ms:u16`.
2. COMMIT within ~2 s: `token:u32`. Expired $\rightarrow$ `TOKEN_EXPIRED`. Wrong token $\rightarrow$ `NOT_ARMED`.
3. After COMMIT wait `min(delay_ms, 5000)` so the host can read the response. 0 is immediate.
4. ACTION_CANCEL cancels before COMMIT.

action	Name	Behavior
1	HOST_LOGICAL_OFF	Enter SHUTDOWN_PENDING
2	HOST_HARD_RESET	EN low 250 ms then high
3	HOST_NORMAL_RESTART	Same as hard reset (BOOT=1)

action	Name	Behavior
4	HOST_ENTER_DOWNLOAD_MODE	BOOT=0 → EN pulse → BOOT=1 after 500 ms
5	HOST_EXIT_DOWNLOAD_MODE	BOOT=1 then hard reset
6	HOST_POWER_ON	Raise EN only from OFF, wake_reason=3
7	HOST_SOFT_SLEEP	RUNNING only

Convenience commands `HOST_SOFT_SLEEP` / `HOST_REQUEST_OFF` / `HOST_REQUEST_RESET` remain available and are still idempotent. Prefer two-phase.

5.2 Cooperative shutdown

- `SHUTDOWN_REQUESTED` (arg0=reason, arg1=10000) or a host off request.
- `power_state` = `SHUTDOWN_PENDING`.
- Host flushes storage and shuts peripherals.
- Send `SHUTDOWN_READY(event_id)`.
- About 100 ms later the PMU drops EN and the slave stops answering.

Forced off (10 s key or `HOST_REQUEST_OFF` with `force=1`) drops EN immediately. At ≤ 3050 mV the PMU starts cooperative shutdown, reason=13.

5.3 Soft sleep

RUNNING only. STATUS becomes `SOFT_SLEEP` immediately. EN drops after ~100 ms.

- While EN is still high, `HOST_READY` cancels the drop and returns to RUNNING.
- `HOST_READY` after EN is already low is **dropped**.
- The next short press only raises EN (wake_reason=2), not a hard-reset pulse.
- Key or alarm power-on clears a pending soft-sleep request.
- DeepSleep is allowed after `SOFT_SLEEP` with EN=0 if idle. An alarm can power the host on.

5.4 IWDT recovery

After a 1 s IWDT reset the PMU powers the host again (wake_reason=4). RTC keeps running if it was already on. Alarms and RAM config are lost.

6. Key behavior

Power key is active low, 30 ms debounce. The PMU drives EN for the rows below. Everything else is events only.

Condition	Action
EN=0, hold ≥ 1 s	Power on. If voltage is valid, ≤ 3050 mV, and not charging: refuse, red blinks twice
EN=0 and <code>SOFT_SLEEP</code> (or <code>RUNNING</code> with EN already low)	Short press raises EN
EN=1, hold ≥ 10 s	Forced power-off
EN=1, two shorts (gap ≤ 1.2 s) then hold ≥ 3 s	Download. The two shorts still report <code>KEY_SHORT</code> ; the third press does not report <code>KEY_LONG</code>
Short / 1.5 s / 8 s	Events only. EN is not changed

`KEY_DOWN` / `KEY_UP` can be disabled with `CONFIG_SET_KEY_EVENTS`. `SHORT` / `LONG` / `FORCE_OFF` stay on.

key_state : bit0 pressed; bit1 past 1.5 s; bit2 past 8 s; bit3 download combo has two shorts.

7. Battery measurement and SOC

7.1 Sampling

1. Enable the divider, wait 10 ms.
2. Multiple samples, drop outliers, average. Also sample the internal bandgap and recover $VDDA = 4095 \times \text{cal} / \text{bandgap_raw}$.
3. $\text{mV} = \text{raw} \times VDDA \times 133 / (4095 \times 100)$. Discard outside 2700–4400 mV.
4. Three failures in a row clear BATTERY_VALID (battery_mv keeps the last good value).
5. On-chip temperature is also reported, 0.01 °C, −40–125 °C.

Scene	Period
Running	~30 s
Off, not charging	~120 s
Off, charging	~60 s
First valid voltage	Within ~800 ms after power-up
Immediate	BATTERY_SAMPLE 0x0100, ACCEPTED, then event 0x11

7.2 Thresholds

Name	Voltage	Behavior
Low	3500 mV	BATTERY_LOW, event 0x12
Critical	3300 mV	BATTERY_CRITICAL, event 0x13
Forced off	3050 mV	Cooperative shutdown reason=13; refuse key power-on
Hysteresis	100 mV	Shared
Full-voltage constant	4150 mV	CONFIG_GET only. Do not use it to decide full

7.3 SOC

Table ends: 4200 mV = 1000‰, 3050 mV = 0‰.

mV	‰
4200	1000
4100	960
4000	900
3900	780
3800	620
3700	450
3650	330
3600	200
3550	110
3450	70

mV	‰
3300	50
3150	20
3050	0

Linear interpolation between points, then smoothing. About ± 10 percentage points, worse on the 3.7–3.8 V plateau. Use for a battery bar. Use threshold events for alarms.

- Not charging: monotonic down. Voltage rebound does not raise the display.
- Charging: hold the pre-charge value; set 1000 when full is inferred.
- Rebase: charge ends, or adjacent samples jump >500 mV (treated as a cell swap).
- Optional low-SOC notify (off by default, 50‰, 30‰ recovery hysteresis).

Equivalent cycle: one count each time accumulated discharge reaches 1000‰. Written to the maintenance page.

8. Charge detection

Use STATUS `charge_state`. Do not decide full from `battery_mv` or `CHARGE_PIN_HIGH` alone.

Value	Name	Rule
0	UNKNOWN	Initial
1	NOT_CHARGING	Continuous low past the adaptive threshold
2	CHARGING	Present and an edge within 2 s
3	FULL_INFERRED	Present and no edge for 3 s
4	FAULT	Reserved

Details:

- Sample period 20 ms; 100 ms when off and no charger.
- Unplug threshold = clamp(last high half-period $\times 1.5 + 200$ ms, 1100, 1800); 1400 ms if none yet.
- For 2.5 s after EN falling, STAT low is not treated as unplug.
- For 2.5 s after EN rising, full is not inferred.
- Changes push `CHARGE_STATE_CHANGED` (arg0=new, arg1=old).
- With `CONFIG_SET_WAKE_ON_CHARGE`, a confirmed plug while the host is OFF powers on, wake_reason=5. Default off, RAM only.

9. LED

Priority high to low:

Priority	Scene	Pattern
1	Download	0.15 s $\times$ 7: off-white-off-red-off-white-off. Not interruptible
2	Shutdown hint	Red blink 150 ms $\times$ 4
3	Power-on self test	Red 150 $\rightarrow$ off 150 $\rightarrow$ white 150 $\rightarrow$ off (every PMU power/reset)
4	Host override	<code>LED_SET</code> (default 5 s), or boot white blink 250 ms / reset red blink 250 ms

Priority	Scene	Pattern
5	Charging	Red breathe, 2.6 s period, 0.3 s dark, starts in the dark segment
5	Full	White solid, 400 ms fade-in. Hidden in BOOT_WAIT
6	Power key held	Red solid
—	EN=0 and not charging	Off (fade out when leaving charge)

`CONFIG_SET_CHARGE_LED=0` disables charge breathe and full-white. `LED_SET`: color 0 off / 1 red / 2 white / 3 both; brightness 0=off, nonzero=on. 2-byte form fades 400 ms; 4-byte form may add `fade_ms`, 0=instant, max 4000 ms.

`led_state`: bit0 red; bit1 white; bit2 red breathe; bit3 host override; bit4 charge policy.

10. Low power

Mode	Condition	Behavior
Normal Sleep	Otherwise	I2C responds
DeepSleep	OFF or <code>SOFT_SLEEP</code> , EN=0, not charging, key idle, LED idle, no pending command, ≥ 15 s since last command and I2C	High-speed clock off, μ A class. RTC keeps time

Wake sources: power-key edge, charger insert, RTC alarm. Invalid wakes go back to sleep immediately.

I2C and DeepSleep:

- No I2C response in DeepSleep.
- After cold start, before the slave has stopped answering: a short press (<1 s) wakes and opens a 15 s window; any I2C activity restarts the timer.
- After logical off or soft sleep the slave stops and releases the bus. Power the host again first (OFF: hold ≥ 1 s; `SOFT_SLEEP`: short press).
- The slave also stops during reset / download pulses and resumes when EN is high again.

Watchdog is paused in DeepSleep. Entry count is `DIAGNOSTICS` offset 14.

11. Time and alarm

- On-chip calendar RTC, LSI clock. Keeps time in DeepSleep and while the host is off.
- `TIME_SYNC` (`0x0005`): UTC seconds, 946684800–4102444799 (2000–01–01 00:00:00 to 2099–12–31 23:59:59). Does not stretch SCL.
- `TIME_GET` (`0x0007`): `unix:u32` (0 if never synced) + `millis:u16` (always 0) + `synced:u8` + reserved.
- After PMU reset `synced=0`. Alarm remaining seconds stay the same when `TIME_SYNC` moves the target.
- `ALARM_SET`: mode 0 off / 1 one-shot / 2 repeat + seconds (10–2678400). Fires `ALARM_FIRED` (arg0=mode). Powers on from OFF or `SOFT_SLEEP`, `wake_reason=6`. RAM only.
- Event `timestamp_ms` is PMU uptime milliseconds, not UTC.
- `RTC_RAW` (`0x000A`) returns 9x32 raw registers for debug.

12. I2C programming interface

Little-endian. Structures are in `pmu_protocol.h` in the same folder.

12.1 Bus

Item	Value
7-bit address	0x2A (write 0x54, read 0x55)
Rate	400 kHz, 100 kHz compatible
Register address	1 byte
Max transfer	1-byte register + 64-byte data
Suggested timeout	20 ms per try; up to 3 tries, backoff 2 / 5 / 20 ms
Stuck recovery	Host may clock up to 9 SCL then STOP; slave self-recovers after ~100 ms with no bus activity

Read (repeated start recommended):

```
START + 0x54 + REG + RESTART + 0x55 + N bytes + STOP
```

W+REG+STOP then R+N+STOP also works.

Write:

```
START + 0x54 + REG + PAYLOAD + STOP
```

12.2 Register map

Addr	Name	Len	Access	Use
0x00	IDENTITY	32	R	Protocol / firmware / boot_id
0x20	STATUS	64	R	Power, battery, charge, key, LED
0x60	DIAGNOSTICS	32	R	Error counters
0x80	COMMAND	64	W	Command frame
0x81	RESPONSE	64	R	Last command response
0x82	EVENT_PEEK	16	R	Head, does not pop. All zeros if empty
0x83	EVENT_ACK	4	W	event_id:u16 + crc16:u16 (CRC over the 2 bytes)
0x84	EVENT_COUNT	1	R	Unacked count
0x85	QUICK_BATTERY	8	R	Battery snapshot

Read a full block and check CRC. Fields in one read are one snapshot.

12.3 IDENTITY (0x00, 32)

Off	Type	Field	Meaning
0	u8[4]	P M U 1	Wrong magic means this is not the chip
4	u8	protocol_major = 1	Must be 1
5	u8	protocol_minor = 1	Must be 1
6	u8	fw_major = 1	Firmware major

Off	Type	Field	Meaning
7	u8	fw_minor = 0	Firmware minor
8	u8	fw_patch = 8	This datasheet is 1.0.8
9	u8	hw_revision = 1	Hardware revision
10	u16	device_id = 0x3201	Must match
12	u32	boot_id	Session id for command session_id . Never 0. Re-read and drop old sequence if it changes
16	u32	capabilities	bits 0–5, see 1.4
20	u8	max_frame_size = 64	Command/response length
21	u8	event_fifo_depth = 8	Event queue depth
22	u16	build_number	Low 16 bits of 20260905
24	u32	build_hash	Must be 20260905
28	u16	reserved	Reads 0
30	u16	crc16 (bytes 0–29)	Must check; discard the block on fail

12.4 STATUS (0x20 , 64)

Off	Type	Field	Host use
0	u16	generation	+1 when the block refreshes
2	u8	power_state	Chapter 5. Use this for command completion
3	u8	wake_reason	Why EN was raised, see 12.11
4	u32	status_flags	Table below
8	u16	battery_mv	Valid only with BATTERY_VALID
10	u16	battery_adc_raw	Divider ADC raw, logging
12	u16	battery_soc_permille	0–1000. Validity is QUICK flags bit1
14	i16	mcu_temp_cent_i_c	On-chip temp, 0.01 °C; 0 = not measured
16	u8	charge_state	Chapter 8. UI uses this only
17	u8	key_state	Chapter 6
18	u8	led_state	Chapter 9
19	u8	host_state	0 unknown 1 off 2 starting 3 ready 4 soft sleep 5 shutting down 6 fault
20	u32	fault_flags	bit0: BOOT_WAIT timeout, sticky until a successful HOST_READY
24	u8	pending_events	Unacked count, same as 0x84
25	u8	cw_reset_reason	Last PMU reset, see 12.11
26	u8	reserved	Reads 0
27	u8	last_command_status	Low 8 bits of last command status
28	u32	uptime_ms	Milliseconds since PMU power-up
32	u32	reserved	Reads 0
36	u32	config_generation	+1 on config change or factory default
40	u16	last_command_sequence	Next command starts at this + 1, especially after USB-only host reset
42	u16	last_event_id	Last queued event id, compare with PEEK

Off	Type	Field	Host use
44	u32	last_action_reason	Last power-action reason, below
48	u8[14]	reserved	Reads 0
62	u16	crc16 (bytes 0–61)	Must check; discard the block on fail

status_flags :

Bit	Name	Meaning
0	BATTERY_VALID	battery_mv is usable
1	BATTERY_LOW	Below 3500 mV, held until hysteresis
2	BATTERY_CRITICAL	Below 3300 mV, held until hysteresis
3	CHARGING_ACTIVE	charge_state is charging
4	CHARGE_PIN_HIGH	STAT was high on this sample; not enough to call full
5	KEY_PRESSED	Power key is down
6	HOST_EN_HIGH	ESP_EN is high
7	HOST_BOOT_LOW	ESP_BOOT is low
8	EVENT_PENDING	Unacked events
9	COMMAND_BUSY	A command is in progress; rare in this firmware
10	CONFIG_VALID	Always 1
13	CW_INT_ASSERTED	CW_INT is low, same time as EVENT_PENDING

last_action_reason : 10 = 10 s key forced off; 11 = cooperative timeout; 12 = SHUTDOWN_READY done; 13 = low-battery cooperative off; 20 = key combo download. Host PREPARE reason is stored as-is.

12.5 QUICK_BATTERY (0x85 , 8)

Off	Type	Field	Meaning
0	u16	battery_mv	Battery voltage; requires flags bit0
2	u16	soc_permille	0–1000; 0xFFFF if invalid
4	u8	charge_state	Same as STATUS, chapter 8
5	u8	flags	bit0 voltage valid, bit1 SOC valid, bit2 charging
6	u16	crc16 (bytes 0–5)	Must check

12.6 DIAGNOSTICS (0x60 , 32)

Counters saturate at 0xFFFF. CLEAR_DIAGNOSTICS zeros the first 30 bytes.

Off	Field	Meaning
0	i2c_rx_errors	Over-long writes and other RX errors
2	i2c_bus_recovery_count	Slave self-recoveries
4	frame_crc_errors	Frame CRC errors
6	bad_length_errors	Length errors
8	unknown_command_errors	Unknown opcodes
10	reserved	Reads 0

Off	Field	Meaning
12	event_overflow_count	Event overflows
14	deepsleep_entry_count	DeepSleep entries
16	host_reset_count	Host hard resets
18	reserved	Reads 0. IWDT reset is STATUS.cw_reset_reason==4
20	adc_error_count	ADC failures
22	reserved	0
24	reserved	0
26	reserved	0
28	reserved	0
30	crc16 (0–29)	

12.7 Command frame (write 0x80, read 0x81)

Fixed 64 bytes.

Off	Type	Request	Response
0	u8	magic 0xA5	Same
1	u8	header_version 1	Same
2	u8	kind 0	kind 1
3	u8	flags	flags
4	u8	protocol_major 1	Same
5	u8	protocol_minor 1	Same
6	u16	sequence, not 0	Echo
8	u16	opcode	Echo
10	u16	0	Status
12	u8	payload length 0–44	Response length
13	u8	reserved	reserved
14	u32	session_id = current boot_id	Echo boot_id
18	u8[44]	payload, pad 0	payload
62	u16	crc16 (0–61)	crc16

CRC-16/CCITT-FALSE: poly= 0x1021, init= 0xFFFF, refin/refout=false, xorout=0. Vector "123456789" → 0x29B1.

```
uint16_t pmu_crc16_ccitt_false(const uint8_t *data, uint16_t length)
{
    uint16_t crc = 0xFFFF;
    while (length--) {
        crc ^= (uint16_t)(*data++) << 8;
        for (uint8_t i = 0; i < 8; ++i) {
            crc = (crc & 0x8000)
                ? (uint16_t)((crc << 1) ^ 0x1021)
                : (uint16_t)(crc << 1);
        }
    }
    return crc;
}
```

Flow: read IDENTITY → fill session_id → write 64 bytes to 0x80 → read 0x81. Async completion may also push COMMAND_COMPLETED.

Idempotency key (session_id, sequence): the same request returns the cached result; same key with different opcode/payload → SEQUENCE_CONFLICT. Cache holds 4. If the response is lost, read again with the same sequence.

Exception: a cached HOST_READY still runs if the current state is not RUNNING (USB-only ESP reset handshake). Off / reset / download do not have this exception.

sequence==0 → INVALID_ARGUMENT. Wrong session_id → STALE_SESSION. Wrong protocol_major or header_version → UNSUPPORTED_VERSION.

Commands finish in place. The host rarely sees BUSY (0x0002). If RESPONSE looks stale, compare sequence.

12.8 Status codes

Value	Name	Meaning
0x0000	OK	Done; response payload is valid
0x0001	ACCEPTED	Accepted; completion is STATUS or an event
0x0002	BUSY	Reserved; unused in this firmware
0x0010	BAD_CRC	Resend the same sequence
0x0011	BAD_MAGIC	Header is not 0xA5
0x0012	BAD_LENGTH	Payload length does not match the command
0x0013	UNSUPPORTED_VERSION	Wrong header_version or protocol_major
0x0014	UNKNOWN_COMMAND	Opcode not supported, including 0x0600–0x06FF
0x0015	INVALID_ARGUMENT	Out of range, or sequence is 0
0x0016	SEQUENCE_CONFLICT	Same number, different content; allocate a new sequence
0x0017	STALE_SESSION	session_id is not current boot_id; re-read IDENTITY
0x0018	NOT_SUPPORTED	Known but unimplemented, e.g. BATTERY_SET_PROFILE
0x0020	INVALID_STATE	Not allowed in this power state, e.g. soft sleep outside RUNNING
0x0021	NOT_ARMED	COMMIT without PREPARE or with a bad token
0x0022	TOKEN_EXPIRED	PREPARE token older than ~2 s
0x0023	PERMISSION_DENIED	Not allowed
0x0030	CONFIG_INVALID	Reserved
0x0031	NVM_FAILURE	Reserved

Value	Name	Meaning
0x0040	ADC_FAILURE	Reserved; sample failures still return ACCEPTED and an event
0x0041	HOST_TIMEOUT	Reserved; BOOT_WAIT timeout only sets <code>fault_flags</code> bit0
0x00FF	INTERNAL_ERROR	Re-read STATUS

12.9 Events

Depth 8, 16 bytes each. `CW_INT` stays low while not empty.

Off	Type	Field	Meaning
0	u16	event_id	From 1, skips 0. ACK must use this value
2	u8	type	Kind, table below
3	u8	severity	0 INFO / 1 NOTICE / 2 WARNING / 3 CRITICAL
4	u32	timestamp_ms	PMU uptime ms, not UTC
8	u32	arg0	Depends on type
12	u16	arg1	Depends on type
14	u16	crc16 (0–13)	Must check

ACK pops the head only; a wrong ID does nothing. When full, oldest INFO is dropped first, then NOTICE; WARNING/CRITICAL may evict lower severity. Drops increment OVERFLOW. Do not use `EVENTS_CLEAR_ALL` as the normal path.

Recommended: `CW_INT` low → read `0x84` → loop PEEK + handle + ACK until count=0. Still read STATUS once per second.

type	Name	Default severity	arg0 / arg1	Host
0x01	KEY_DOWN	INFO	0 / 0	Optional; can disable
0x02	KEY_UP	INFO	hold ms / 0	Optional
0x03	KEY_SHORT	NOTICE	duration / 0	Host decides
0x04	KEY_LONG	NOTICE	duration / 0	Host decides
0x05	KEY_FORCE_OFF	WARNING	duration / 0	Host decides; PMU does not cut power
0x10	CHARGE_STATE_CHANGED	NOTICE	new / old	Update UI
0x11	BATTERY_SAMPLE_READY	INFO	mV / raw	Update voltage
0x12	BATTERY_LOW	WARNING	mV / threshold	Alert
0x13	BATTERY_CRITICAL	CRITICAL	mV / threshold	Alert; ≤3050 starts cooperative shutdown
0x14	BATTERY_SOC_LOW	WARNING	SOC‰ / threshold‰	Must enable first
0x15	ALARM_FIRED	NOTICE	mode / 0	Host already powered on if it was off
0x20	HOST_STARTED	NOTICE	wake_reason / 0	Send HOST_READY soon
0x21	HOST_READY	NOTICE	boot ms / boot_reason	Echo
0x22	SHUTDOWN_REQUESTED	WARNING	reason / deadline_ms	Flush, then SHUTDOWN_READY
0x24	HOST_RESET_PERFORMED	NOTICE	reason / count	Host was hard-reset; log if needed
0x30	COMMAND_COMPLETED	INFO	sequence / status	Match sequence for async commands

type	Name	Default severity	arg0 / arg1	Host
0x40	CONFIG_RECOVERED	NOTICE	generation / reason	Defaults restored; re-read CONFIG_GET
0x41	CW_RESET	WARNING	reset_reason / boot_id low 16	Rebuild the session
0x7F	OVERFLOW	CRITICAL	lost_count / 0	FIFO dropped events

12.10 Commands

System

Opcode	Name	Request payload	Response	Note
0x0001	PING	empty	empty	Probe
0x0002	GET_EXTENDED_INFO	empty	6 B: fw 3 bytes + hw + build_number u16	Check firmware and build
0x0004	HOST_READY	boot_reason:u8	empty	Required after boot. From BOOT_WAIT / POWERING_ON / RESETTING / RUNNING / DOWNLOAD_MODE / (SOFT_SLEEP while EN still high) → RUNNING. Ignored in SOFT_SLEEP after EN is low
0x0005	TIME_SYNC	unix:u32	empty	UTC seconds, chapter 11
0x0006	CLEAR_DIAGNOSTICS	optional	empty	Clears first 30 DIAG bytes; mask ignored
0x0007	TIME_GET	empty	8 B	Current UTC, chapter 11
0x0008	ALARM_SET	mode:u8 + seconds:u32	empty	Off / one-shot / repeat, chapter 11
0x0009	ALARM_GET	empty	10 B: mode + rsvd + remaining + target_unix	Remaining seconds and target
0x000A	RTC_RAW	empty	36 B: 9×u32	Raw calendar registers

If HOST_READY and HOST_SOFT_SLEEP are sent together, READY finishes first.

Battery and LED

Opcode	Name	payload	Note
0x0100	BATTERY_SAMPLE	flags:u8 ignored	Immediate sample, ACCEPTED, then event 0x11
0x0101	BATTERY_SET_PROFILE	—	NOT_SUPPORTED
0x0200	LED_SET	color + brightness [+ fade_ms]	Temporary LED override, chapter 9
0x0203	LED_OVERRIDE_CLEAR	empty	End LED_SET early and fade out

Power

See chapter 5. Also:

Opcode	Name	payload	Note
0x0303	SHUTDOWN_READY	request_event_id:u16	Reply after flush, using the SHUTDOWN_REQUESTED event_id

Opcode	Name	payload	Note
0x0310	HOST_SOFT_SLEEP	empty	RUNNING only. Otherwise INVALID_STATE
0x0311	HOST_REQUEST_OFF	force:u8 + reason:u16	force=0 cooperative; force=1 immediate off
0x0312	HOST_REQUEST_RESET	reason:u16	Hard-reset the host
0x0313	EVENTS_CLEAR_ALL	empty	Clears the FIFO. Do not use day-to-day; ACK one by one

Config (RAM only; defaults after reset)

Opcode	Name	payload	Default
0x0500	CONFIG_GET	empty	36 B, below
0x0501	CONFIG_SET_WAKE_ON_CHARGE	enable:u8	0
0x0502	CONFIG_SET_LOW_SOC_NOTIFY	enable:u8 [+ threshold:u16], 1–1000	off, 50%
0x0503	CONFIG_SET_KEY_EVENTS	enable:u8	1
0x0504	CONFIG_SET_CHARGE_LED	enable:u8	1
0x0505	CONFIG_FACTORY_DEFAULT	empty	Restores the RAM switches above; does not change the maintenance page or VCOM

0x0600–0x06FF is always **UNKNOWN_COMMAND** (no IAP).

CONFIG_GET 36 bytes:

Off	Type	Field	Value
0	u16	low_battery_mv	3500
2	u16	critical_battery_mv	3300
4	u16	key_long_press_ms	1500
6	u16	key_force_event_ms	8000
8	u16	key_power_on_ms	1000
10	u16	key_force_off_ms	10000
12	u16	full_battery_mv	4150 (do not use as full)
14	u32	config_generation	+1 on config change or factory default
18	u16	forced_off_mv	3050
20	u16	hysteresis_mv	100
22	u16	key_debounce_ms	30
24	u16	host_boot_timeout_ms	15000
26	u16	shutdown_timeout_ms	10000
28	u16	led_override_ms	5000
30	u8	wake_on_charge	Current, default 0
31	u8	key_raw_events	Current, default 1 (KEY_DOWN/UP)
32	u8	charge_led_enabled	Current, default 1
33	u8	low_soc_notify_enabled	Current, default 0
34	u16	low_soc_threshold_permille	Current, default 50

VCOM

Opcode	Name	payload	Note
0x0510	VCOM_GET	empty	4 B: vcom_mv:u16 + valid:u8 + 0. Uncalibrated valid=0, mv=0
0x0511	VCOM_SET	vcom_mv:u16	500–2500, 10 mV aligned. Stored immediately. Echo 2 B

Unit is absolute mV (1290 = −1.29 V). Illegal values return INVALID_ARGUMENT . Flash write takes a few milliseconds.

Device info

MAINT_GET_INFO (0x05F2). 36–byte response:

Off	Type	Field	Meaning
0	u8[10]	Factory UID	Silicon UID, usable as serial
10	u16	Equivalent cycles	Full–discharge counts
12	u16	Battery replace count	Service–written replace count
14	u16	Cycle accumulator %	0–999; carry at 1000
16	u8[16]	Repair codes	0x00 = empty slot
32	u8	flash_readout_level	Current readout–lock level
33	u8	Readout–lock change count	Times changed
34	u8	Remaining	Remaining changes
35	u8	reserved	0

12.11 Reset / wake reasons

cw_reset_reason : 0 unknown, 1 POR, 2 brown–out, 3 NRST, 4 IWDT, 5 software, 6 LOCKUP, 7 clock. This firmware reports: IWDT → 4, POR → 1, NRST → 3, else 0.

wake_reason : 0 unknown, 1 cold–start off, 2 key on, 3 host request on, 4 IWDT recovery, 5 charger insert (must be enabled), 6 RTC alarm.

13. Persistence and VCOM

Flash page 127 (0xFE00–0xFFFF , 512 B) holds a 32–byte record. Each update erases and writes the page. Endurance ≥20k cycles. The CPU stalls a few milliseconds during erase/write.

Off	Type	Field
0	u32	magic PVM1 (0x314D5650)
4	u16	version = 1
6	u16	cycle_count
8	u16	replace_count
10	u16	cycle_accum_permille
12	u8[16]	repair_codes
28	u16	vcom_mv (0 / 0xFFFF = uncalibrated)
30	u16	crc16 (0–29)

Written on cycle carry, VCOM write, and logical shutdown. `CONFIG_FACTORY_DEFAULT` restores RAM switches only.

Host: if NVS is valid, use NVS and write it back to the PMU when possible; if NVS is invalid, `VCOM_GET`; if still invalid, show calibration UI and `VCOM_SET` after confirm.

14. Production, programming, and readout lock

Item	Value
Toolchain	Keil MDK / ARMCC
Image	<code>read_pico_pmu_v1.0.8_build20260905.hex</code>
Programming	SWD (PA07 / PA08)
Power-on test	Red then white blink, 0.6 s total
Readout lock	ISP only: CW_Programmer + CW-Writer (or USB-UART with RTS), level 1

After lock, SWD/ISP cannot read the firmware. Downgrade requires a full-chip erase and clears the maintenance page.

15. Reliability

- Replay protection: commands must carry the current `boot_id`.
- Idempotency for dangerous actions.
- CRC16 on every structured block.
- IWDT 1 s, paused in DeepSleep, restores host power after reset.
- I2C slave self-recovers after ~100 ms on a stuck bus.
- Low battery refuses power-on, avoiding boot/brown-out loops.

Appendix A Acceptance checklist

1. `0x2A` reads IDENTITY/STATUS/DIAGNOSTICS with good CRC.
 2. IDENTITY: `PMU1`, protocol 1.1, firmware 1.0.8, `build_hash==20260905`.
 3. After `boot_id` changes, old commands return `STALE_SESSION`.
 4. Short / long / 8 s are events only; 10 s forces power-off.
 5. Plug → `charge_state==2`, full → `==3`, unplug → `==1`; do not override from voltage.
 6. After `BATTERY_SAMPLE`, event `0x11` or a new `battery_mv`.
 7. Events must be ACKed; a full FIFO overflows.
 8. Send `HOST_READY` after boot; flush then `SHUTDOWN_READY` on cooperative shutdown.
 9. While charging, `LED_SET` may be covered by the charge LED unless that LED is disabled.
 10. Blank part: `VCOM_GET` → `mv=0`, `valid=0`; `SET 1290` survives power loss; illegal values → `INVALID_ARGUMENT`.
 11. `CONFIG_FACTORY_DEFAULT` restores RAM switches only; VCOM stays.
 12. Soft sleep: STATUS becomes `SOFT_SLEEP` first, `EN=0` after ~100 ms; short press only raises EN.
 13. Time sync does not stretch SCL; I2C still works afterwards.
-

Appendix B Document information

This document covers read_pico_pmu firmware 1.0.8, protocol 1.1, on Read Pico. Same folder: pmu_protocol_en.h and pmu_registers_en.json.

Item	Value
Company	Shenzhen MindReset Technology Co., Ltd.
Date	2026-09-17
Firmware	1.0.8
Protocol	1.1
Build fingerprint	20260905